

Solidity and Blockchain

Understanding and implementing a smart contract

Priyanka Srinivas | feedback@digit.in

PREREQUISITES

To learn the principle behind blockchain and smart contracts, there is some software required such as Node, Node Package Manager (npm), VSCode or Sublime Text. You must also have basic knowledge of HTML, CSS and specifically intermediate JavaScript principles. With all that in hand, you're ready to take your first shot at building a smart contract!

WHAT IS A SMART CONTRACT?

If you have heard of decentralised apps (or DApps) before, it'll be easier to understand a smart contract. A DApp is an application that is launched on a decentralised network. Just like regular applications are made from some modules of code, a DApp is a set of multiple smart contracts.

A smart contract is a set of self-executable instructions written in code instead of a pen and paper. These codes are hosted, or made live, on a blockchain. A smart contract allows the people participating in the contract to agree to certain terms that can be executed without a third-party or external governing force. Smart contracts are usually built on a blockchain network called ethereum.

Ethereum finds its base in bitcoin, which is a digital currency used to make transactions online. Since bitcoin is not owned by any agency, it is known

as 'decentralised', which extends to any protocols built on it such as ethereum.

GETTING STARTED: DEPENDENCIES

Ganache

Ganache is used for setting up a personal Ethereum Blockchain for testing your Solidity contracts. Before you begin using Ganache, you must first download and install the Blockchain on your local machine. You can do so by going to: <https://truffleframework.com/ganache>

Ganache is available on several platforms. When you open the installation URL given above, it automatically detects your machine's OS and directs you to the appropriate binary installation. After installing, the screen of the application will look like this:



Clicking on QUICKSTART will start the application and lead you to the Ganache console where you may see different accounts along with their balance and transaction count. Ganache comes with test accounts that contain 100 ETH each, to purchase deployment gas for smart contracts. By default, each transaction cost will be deducted from the first available Ganache test account. We will be working with the networks

provided by Ganache. Hence, we will be working on a local blockchain.

Truffle

It is a framework to develop smart contracts using solidity language. Please make sure you have Node installed, because Node Package Manager is essential to manage frameworks used to create web apps on a centralised or decentralised network. Use the following command to install truffle: `npm install -g truffle@5.0.2`

INTRODUCTION TO THE BLOCKCHAIN

To learn any new framework, a to-do list app is the best way to test multiple aspects of that framework, mainly CRUD functions. A blockchain to-do list can be accessed via a browser. There needs to be a client-side application that a programmer would build themselves. Traditionally, this client-side application would talk to a web backend and database hosted on a central network that would then fetch data and display according to the request being made by the client.

However, in a blockchain to-do list, the client-side application will directly talk to a blockchain. On this blockchain, you can host code that will be written using ethereum smart contracts that will contain the logic of the to-do list. All the to-do list items will be stored on the blockchain itself. We will not be creating an entire to-do list app. However, we will go through the setup of any blockchain app, and deploy our first smart contract. This smart contract will contain a variable to count to the total number of tasks in the to-do list.

SETTING UP THE PROJECT

Firstly, create a directory for your project. Check the truffle version. Launch the truffle framework for the project using the command `truffle init`

Your project is initialised!

Now, every project needs some package dependencies for different cases such as server, frontend etc. Truffle does not contain a `package .`